

Firza Chandra Sandjaya Putra

Jakarta, Indonesia | +6285156983459 | firzasandjaya@gmail.com

Website for detailed Portfolio & Project: <https://firzacank.vercel.app/>

<https://linkedin.com/in/firzaputra/> | <https://github.com/FirzaCank> | <https://firzachandra.my.canva.site/> |

<https://hackerrank.com/profile/firzasandjaya>

Data Engineer, ML Engineer, and AI Engineer with ~4 years of expertise in building end-to-end data pipelines & scalable MLOps frameworks. I focus on the evolution of built data infrastructure/warehouse into intelligent automation, specializing in the deployment and optimization of ML models that accelerate workflows. Proven track record in Telco and Consumer brands of delivering high-impact systems that drive automated, data-led decisions.

WORK EXPERIENCE

PT Hypefast Karya Nusantara – Jakarta, Indonesia

March 2025 - Now

Data Engineer (Machine Learning OPS)

- **Designed L0–L2 MLOps and ETL pipelines** ingesting order and customer data from 4 marketplace APIs & Google Sheets to BigQuery for **11+ brands, with Flask-based token management** on Cloud Run and PostgreSQL for API authentication, enabling unified analytics and reporting.
- Optimized scalable **product recommendation systems on Shopee Marketplace** (NCF, Replenishment, and Baby Cloth) with full MLOps on GCP: Vertex AI Training, Model Registry, Experiments, and Endpoints, served via FastAPI on Cloud Run, **driving approximately IDR 300M/month GMV**.
- Created an end-to-end pipeline for a **smart voucher optimization system** using K-Means clustering and linear programming for budget-constrained allocation, orchestrated via Vertex AI Pipelines and Cloud Composer, **achieving 200× average monthly ROI per brand**.
- **Automated financial workflows** for 3 brands by integrating Jubelio (OMS) with Jurnal.id via API, **reducing invoice processing from days to <5 minutes** for hundreds of daily invoices, and **accelerating cash flow** through faster AR recognition.
- Engineered an **AI-based OCR** system using GCP Document AI (evaluated against Mistral AI), automating invoice & receipt processing, **reducing extraction time from hours to 5 minutes (96× faster)** with 90%+ accuracy.
- **Delivered Metabase dashboards** for Finance, Data Science, and Business Operations teams to monitor catalog performance, advertising AI, and customer data platform.

PT IDstar Cipta Teknologi (Placement in PT XL Axiata Tbk.) – Jakarta, Indonesia

July 2022 - March 2025

Data Engineer and Analytics / ETL Developer

- Developed an end-to-end telco user behavior & credit scoring platform using a Medallion Architecture (Bronze/Silver/Gold) on Snowflake; built PySpark ETL pipelines on AWS (S3, EC2, Lambda) **processing 1B+ daily records with schema evolution**, partition pruning, and data quality gates enforced at each layer.
- Built a monthly PySpark-based lead generation pipeline to process and **enrich millions of XL Axiata subscriber profiles** for targeted bank product acquisition.
- Designed a dashboard to monitor XL's API Economy performance for external partners, **improving API response times by 30%, success rates by 15%, and driving 10% revenue growth from API transactions**.

Telkom Indonesia – Jakarta, Indonesia

February 2021 - August 2021

Data Scientist Internship

- Support data cleansing, analyzed an exploratory dataset, reported, and interpreted the results to drive insights and data-driven decision-making on T-Money within 1 month.
- Developed a business model for new digital products, KALISA, a waqf blockchain system, and the result was implemented into the new KALISA business model within three months.

PROJECT PORTFOLIO AND SKILLS

Freelance & Projects:

- **Social Media Scraping Integration** - Integrations between 5 platforms: Facebook, X, Instagram, TikTok, and YouTube for Marketing Agency research purposes with ETL using MongoDB.
- **Google News Scraping and Parsing** - Automate extraction, integrate GEMINI AI, DIFFBOT, and APIFY for parsing, and aggregate news articles for sentiment and trend analysis for Marketing Agency research.

- **Debt Collection Performance Analytics** - Built an end-to-end pipeline and Looker Studio, identified critical bottlenecks with actionable recommendations
- **Writer Density Index (WDI)** - Quantitative Analyst Developed a PCA-based composite model (68.2% variance explained) to analyze regional writing ecosystem strength across Indonesia, delivering policy-ready insights for strategic decision-making.
- **Strategic Valuation & Growth Outlook of Blibli Pitch Deck** - Built an end-to-end process delivering strategic analysis, financial forecasting, and DCF & comparable valuation to support investment decision-making and highlight re-rating potential through profitability improvement.
- **Cylinder Lifetime & Rehouse Monitoring Dashboard** - Built a preventive maintenance dashboard for excavator cylinders, tracking lifetime status, rehousing timeline, and stock vs demand analysis to optimize scheduling and reduce downtime.
- **Discourse Network Analysis (DNA) and Social Network Analysis (SNA)** - Built a multi-platform pipeline DNA and SNA with pyvis and Chart.js interactive dashboard, parsing with Gemini LLM to analyze Indonesian nuclear energy policy discourse.
- **Affiliate Commission Performance Review Analysis** - Diagnosed a 34% monthly revenue decline by analyzing conversion funnel leaks and cohort retention collapse (23% to 9%), delivering a strategic recovery roadmap to reclaim Rp51.8M.
- **Mobile Analytics Dashboard Monitoring & AI Insight Pipeline Digital Gold Financial Services App** - Built an end-to-end data pipeline integrating 4 sources (Apple App Store Connect, Google Play Console, AppsFlyer, and MoEngage) into BigQuery via dbt-modeled staging/mart layers, orchestrated by daily Cloud Run Jobs and Workflows, powering a real-time dashboard with AI-driven insights across acquisition, engagement, and retention.

Tech Stack and Technical Skills/Competencies:

- **AI/LLM:** LangChain, LangGraph, OpenAI API, Anthropic Claude API, Gemini API, FAISS, RAG, SentenceTransformer, OpenCV
- **ML/Data Science:** TensorFlow/Keras, PyTorch, scikit-learn, pandas, numpy, PySpark
- **Backend:** FastAPI, Flask, SQLAlchemy, REST API
- **On-Premise:** Cloudera (CDH), Apache Airflow, Kafka, Hive, Impala, Hadoop (HDFS, YARN)
- **Cloud:** AWS: EC2, S3, Lambda, ECS/ECR, Snowflake. GCP: Vertex AI & Agent Platform (Training, Endpoints, Model Registry, Experiments), BigQuery, Cloud Run, Storage, Pub/Sub, DocumentAI, Composer.
- **Databases:** PostgreSQL, MySQL, MongoDB Atlas
- **Infra & DevOps:** Docker, Kubernetes, BitBucket CI/CD, Linux, Git
- **BI & Visualization:** Metabase, Looker Studio, Power BI, Tableau, Library seaborn & matplotlib
- **Tools:** Google Sheets, Apps Script, Jira, Confluence, N8N, dbt, Bitwarden
- **Programming Language:** Python, SQL, JavaScript

EDUCATION

Bandung Institute of Technology (ITB), Bandung

Bachelor's Degree - Industrial Engineering

- **Paid assistant of the laboratory:** Production System Laboratory (LSP) on Faculty of Industrial Technology

LEADERSHIP EXPERIENCE

INFINITY ITB (ITB Dance and Performance Art Community) – ITB

Vice President and Head of the General Secretary Department

- Led organizational strategy, including vision, mission, and OKRs, overseeing budgeting and external partnerships - achieving 95% program execution success across all divisions.
- Led the General Secretariat Department, managing three divisions: Treasury, Administration, and Secretary Management.

Keluarga Mahasiswa Teknik Industri (MTI ITB) – ITB

Head of Staff of People Management & Measurement

- Performed the Social Mapping project for MTI 2018 students to identify the 10 best group combinations.